



EXERCISE



Multiple Choice Questions

Choose the correct answer

MARK
1

Based on Unit and Dimension

1. X , Y and Z are three physical quantities having units $\text{g} \cdot \text{cm}^2 \cdot \text{s}^{-5}$, $\text{g} \cdot \text{s}^{-1}$ and $\text{cm} \cdot \text{s}^{-2}$. Relationship between X , Y and Z can be represented by

- (A) $X \propto YZ$ (B) $X \propto YZ^2$ (C) $X \propto Y^2Z$ (D) $X \propto Y/Z$ (B)

2. Steradian is the unit of

- (A) angle (B) solid angle
(C) arc of a circle (D) circumference (B)

3. Dimension of electric potential is the same as that of

- (A) current (B) force
(C) electromotive force (D) energy (C)

4. The dimensional formula of potential difference is

- (A) $\text{ML}^2\text{T}^{-3}\text{A}^{-1}$ (B) $\text{MLT}^{-3}\text{A}^{-1}$
(C) $\text{ML}^3\text{T}^{-3}\text{A}$ (D) $\text{ML}^2\text{T}^{-3}\text{A}^{-2}$ (A)

5. Which of the following is dimensionally correct?

- (A) Pressure = energy per unit volume
- (B) Pressure = energy per unit area
- (C) Pressure = force per unit length
- (D) Pressure = force per unit volume

6. The unit of specific conductivity is

- (A) $\Omega \cdot \text{cm}^{-1}$
- (B) $\Omega^{-1} \cdot \text{cm}^{-1}$
- (C) $\Omega \cdot \text{cm}^{-2}$
- (D) $\Omega^{-1} \cdot \text{cm}$

[Manipal 09]

7. As per quantum theory, the energy E of a photon is related to its frequency ν as $E = h\nu$, where $h\nu =$ Planck's constant. Then, the dimension of h would be

- (A) ML^2T^{-2}
- (B) ML^2T^{-1}
- (C) MLT^{-2}
- (D) MLT^{-1}

8. Which of the following quantities is dimensionless, if $v =$ speed, $r =$ radius of a circular path and $g =$ acceleration due to gravity?

- (A) $\frac{v^2 r}{g}$
- (B) $\frac{v^2}{rg}$
- (C) $\frac{v^2 g}{r}$
- (D) $v^2 rg$

Based on Dimensional Analysis

9. Considering force (F), length (L) and time (T) to be fundamental physical quantities, find the dimension of mass.

- (A) FL^{-1}T^2
- (B) $\text{FL}^{-1}\text{T}^{-1}$
- (C) FLT^{-2}
- (D) $\text{F}^{-1}\text{L}^{-1}\text{T}^2$

10. Time period of a pendulum (T), its length (l), mass of its bob (m) and acceleration due to gravity (g) are related as $T = km^x l^y g^z$ where,

- (A) $x = 1, y = \frac{1}{2}, z = \frac{1}{2}$
- (B) $x = 0, y = -\frac{1}{2}, z = \frac{1}{2}$
- (C) $x = 1, y = -\frac{1}{2}, z = \frac{1}{2}$
- (D) $x = 0, y = \frac{1}{2}, z = -\frac{1}{2}$

11. The equation of state of a gas is given by $(p + \frac{a}{V^3})(V - b^2) = cT$, where p , V and T are pressure, volume and temperature respectively and a , b , c are constants. The dimensions of a and b are respectively

- (A) ML^8T^{-2} and $\text{L}^{3/2}$
- (B) ML^5T^{-2} and L^3
- (C) ML^5T^{-2} and L^6
- (D) ML^6T^{-2} and $\text{L}^{3/2}$

[WBJEE '13]

12. The unit of t is s and that of x is m in the expression, $y = A \cos(\frac{t}{p} - qx)$. Then

- (A) x and q have the same unit
- (B) x and p have the same unit
- (C) t and q have the same unit
- (D) t and p have the same unit

13. The unit of both x and y is m in the expression $y = A \sin[\frac{2\pi}{\lambda}(ct - x)]$. Then

- (A) x , λ and A have the same unit
- (B) x and λ have the same unit, but the unit of A is different
- (C) c and $\frac{2\pi}{\lambda}$ have the same unit
- (D) $(ct - x)$ and $\frac{2\pi}{\lambda}$ have the same unit

14. The dimension of ω in the expression $y = A \sin(\omega t - kx)$ is

- (A) M^0LT
- (B) $\text{M}^0\text{L}^{-1}\text{T}^0$
- (C) $\text{M}^0\text{L}^0\text{T}^{-1}$
- (D) M^0LT^{-1}

Based on Errors in Measurement

15. If the error in the measurement of momentum of a particle is 50% then the error in the measurement of kinetic energy is

- (A) 75%
- (B) 100%
- (C) 125%
- (D) 200%

16. Percentage errors in the measurement of mass and speed are 2% and 3% respectively. The error in the estimation of kinetic energy obtained by measuring mass and speed will be

- (A) 12%
- (B) 8%
- (C) 10%
- (D) 2%

[AIIMS '09]

17. Choose the incorrect statement out of the following.

- (A) Every measurement by any measuring instrument has some errors
- (B) Every calculated physical quantity based on measured values has some error
- (C) A measurement can have more accuracy but less precision and vice versa
- (D) The percentage error is different from relative error

[AMU '10]

18. A screw gauge has a pitch of 0.5 mm, and has 50 divisions on its circular scale. Its linear and circular



scale readings are, respectively, 2 mm and 2 s when the diameter of a sphere is measured by the gauge. The result of this measurement is

- (A) 1.2 mm (B) 1.25 mm
(C) 2.20 mm (D) 2.25 mm

19. The smallest main scale division of a vernier caliper = 1 mm. Here, 20 vernier divisions = 16 main scale divisions. The vernier constant is

- (A) 0.02 mm (B) 0.05 mm
(C) 0.1 mm (D) 0.2 mm

20. The measured values of mass, radius and length of a wire are, respectively, (0.3 ± 0.003) g, (0.5 ± 0.005) mm and (6 ± 0.06) cm. The maximum percentage error in the computed value of the density of the material of the wire would be

- (A) 1 (B) 2 (C) 3 (D) 4

Hint: Fractional errors : $\frac{\Delta m}{m} = \frac{0.003}{0.3} = \frac{1}{100}$;

Similarly, $\frac{\Delta r}{r} = \frac{\Delta l}{l} = \frac{1}{100}$

Density (D) = $\frac{m}{\pi r^2 l}$;

So, $\frac{\Delta D}{D} = \frac{\Delta m}{m} + 2\frac{\Delta r}{r} + \frac{\Delta l}{l} = \frac{1}{100} + \frac{2}{100} + \frac{1}{100} = \frac{4}{100}$
= 4%

21. The dimension of a quantity is $M^a L^b T^{-c}$. If the measurements of mass, length and time involve errors of $\alpha\%$, $\beta\%$ and $\gamma\%$, respectively, then the error in the measurement of the given quantity is

- (A) $(\alpha a - \beta b + \gamma c)\%$ (B) $(\alpha a + \beta b + \gamma c)\%$
(C) $(\alpha a + \beta b - \gamma c)\%$ (D) $(\alpha a - \beta b - \gamma c)\%$

Based on Significant Figures

22. What is the number of significant figure in $(3.10 + 4.60) \times 10^7$?

- (A) 5 (B) 3 (C) 4 (D) 7

23. The sum of 6.75×10^3 cm and 4.52×10^2 cm, keeping the significant digits only, is

- (A) 7.202×10^3 cm (B) 72.0×10^2 cm
(C) 0.72×10^4 cm (D) 7.20×10^2 cm

[AIIMS '09]

24. A rectangle has a length = 4.234 m and breadth = 1.05 m. What is its area in m^2 ? Keep the significant digits only.

- (A) 4.4457 (B) 4.45 (C) 4.446 (D) 4.44



Very Short Answer Type Questions

Answer in one or two words

MARK
1

Based on Unit and Dimension

- Write the number of base units in SI. [seven]
- Is mole a base unit or a derived unit in SI? [base]
- What is the unit of thermal capacity in SI? [$J \cdot K^{-1}$]
- Express one parsec in terms of light year.

[1 parsec = 3.26 light year]

- Light year is a _____ unit. [Fill in the blanks]
[fundamental]
- Ampere is a _____ unit in SI. [Fill in the blanks] [base]
- Candela is a _____ unit in SI. [Fill in the blanks] [base]
- Parsec is a _____ unit. [Fill in the blanks] [fundamental]
- What is the dimension of a dimensionless physical quantity? [1]

10. The relative density of lead is 11.3. What is its density in CGS and SI? [NCERT] [$11.3 \text{ g} \cdot \text{cm}^{-3}$, $1.13 \times 10^4 \text{ kg} \cdot \text{m}^{-3}$]

11. (a) $1 \text{ kg} \cdot \text{m}^2 \cdot \text{s}^{-2} = \text{_____} \text{ g} \cdot \text{cm}^2 \cdot \text{s}^{-2}$.
[Fill in the blanks] [10⁷]

- (b) $1 \text{ m} = \text{_____} \text{ light year}$. [Fill in the blanks] [10⁻¹⁶]

- (c) $3.0 \text{ m} \cdot \text{s}^{-2} = \text{_____} \text{ km} \cdot \text{h}^{-2}$. [Fill in the blanks]
[3.9 × 10⁴]

- (d) $G = 6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2 \cdot \text{kg}^{-2} = \text{_____}$
 $\text{cm}^3 \cdot \text{s}^{-2} \cdot \text{g}^{-1}$. [Fill in the blanks]

[NCERT] [6.67 × 10⁻⁸]

Based on Dimensional Analysis

12. If $x = a + bt + ct^2$, where x is in metres and t in seconds, what are the dimensions of b and c ?

[LT⁻¹ and LT⁻²]

13. The equation of state of a real gas is $(p + \frac{a}{V^2})(V - b) = RT$, where p , V and T are pressure, volume and absolute temperature, respectively. Find out the dimension of b . [L³]

Based on Significant Figures

14. The Avogadro number is 6.022×10^{23} . How many significant figures are there? [WBCHSE Sample Question '13]

[4 figures: 6, 0, 2, 2]